Fluids and Lubricants Specifications

Diesel engine-generator sets
with mtu Series 1600 engines

A001068/04E

A Rolls-Royce solution

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1 Preface

1.1 General information

Used symbols and means of representation

The following instructions are highlighted in the text and must be observed:

Important
This field contains product information which is important or useful for the user. It refers to instructions, work and activities that have to be observed to prevent damage or destruction to the material.

Note:

A note provides special instructions that must be observed when performing a task.

Fluids and lubricants

The operational life, operational reliability and function of the drive systems are largely dependent on the fluids and lubricants employed. The correct selection and treatment of these fluids and lubricants are therefore extremely important.

Test standard	Designation
DIN	Federal German Standards Institute
EN	European Standards
ISO	International Standards Organization
ASTM	American Society for Testing and Materials
IP	Institute of Petroleum
DVGW	German Gas and Water Industry Association

Table 1: Test standards for fluids and lubricants

Applicability of this document

These Fluids and Lubricants Specifications apply to fluids and lubricants for diesel engine-generator sets with the following mtu engines:
- Series 1600Gx0

Note: References to other engine series in this document should be disregarded.

Applicability of this publication

The Fluids and Lubricants Specifications will be amended or supplemented as necessary. Before using them, make sure you have the latest version (publication number A001068/..). The latest version is available at: http://www.mtu-solutions.com.

If you have any questions, your contact person will be happy to help you.

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Warranty

Use of the approved fluids and lubricants, either under the brand name or in accordance with the specifications given in this publication, constitutes part of the warranty conditions.

The supplier of the fluids and lubricants is responsible for the worldwide standard quality of the named products.

Important
Fluids and lubricants for diesel engine-generator sets can be hazardous materials. Certain regulations must be obeyed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, legal requirements and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for fluids and lubricants is therefore not possible within this publication.

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. Rolls-Royce Solutions accepts no liability whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

Rolls-Royce Solutions recommends consultation with the suppliers of all fluids and lubricants to request the relevant safety data sheets prior to storing, handling and using these fluids and lubricants.

Safe disposal

Important
To prevent environmental pollution and infringements of statutory requirements, used fluids and lubricants must be disposed of in accordance with local regulations. Never dispose of or burn the used oil in the fuel tank.

The regulations for the disposal of fluids and lubricants differs from place to place. Environmental protection is one of the fundamental corporate objectives of Rolls-Royce Solutions. We therefore recommend the recycling of fluids and lubricants wherever possible. If recycling is not available, Rolls-Royce Solutions recommends contacting the local waste disposal authority, before disposing of any fluids and lubricants to determine the best option. Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. Rolls-Royce Solutions accepts no liability whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

Registered trademarks

All brand names are registered trademarks of the manufacturer concerned.

Preservation

All information on preservation, represervation and depreservation including the approved preservatives is available in the Preservation and Represervation Specifications (publication number A001070/...). The most recent version can be consulted under:

http://www.mtu-solutions.com

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2 Engine Oils

2.1 Requirements and oil change intervals

Important
Dispose of used fluids and lubricants in accordance with local regulations. Used oil must never be disposed of via the fuel tank!

Requirements for the approval of engine oils

The conditions for the approval of engine oils for diesel engines are specified in the delivery standard MTL 5044, which can be ordered under this reference number.

Manufacturers of engine oils are notified in writing if their product is approved.

Diesel engine oils approved for Series 1600 engines are divided into the following quality categories:
- Oil category 2: Higher quality oils / multi-grade oils
- Oil category 2.1: Multi-grade oils with a low ash-forming additive content (low SAPS oils)
- Oil category 3: Highest quality / Multi-grade oils
- Oil category 3.1: Multi-grade oils with a low ash-forming additive content (low SAPS oils)

Low SAPS oils are oils with a low sulfur and phosphor content and an ash-forming additive content of ≤ 1 %. They are only approved if the sulfur content in the fuel does not exceed 50 mg/kg. Depending on the exhaust gas aftertreatment used, the use of low-ash oils is prescribed (→ Page 9).

Selection of a suitable engine oil is based on fuel quality, projected oil drain interval and on-site climatic conditions. At present there is no international industrial standard which alone takes into account all these criteria.

Important
The use of engine oils not approved by Rolls-Royce Solutions can result in increased wear and can mean that statutory emission limits can no longer be observed. This can be a punishable offense.

Special features of Rolls-Royce Solutions diesel engine oils

The following multi-grade oils are available depending on the relevant region.

Multi-grade oils from Rolls-Royce Solutions

Manufacturer & sales region	Product name	SAE grade	Oil category	Material no.
Rolls-Royce Solutions GmbH Europe Middle East Africa	Diesel Engine Oil DEO SAE 15W-40	15W-40	2	20 l canister: X00070830 210 l barrel: X00070832 IBC: X00070833 Loose items: X00070835 (only on request)
Rolls-Royce Solutions America Inc. Americas	Power Guard® SAE 15W-40 Off Highway Heavy Duty	15W-40	2.1	5 gallons: 800133 55 gallons: 800134 IBC: 800135
Rolls-Royce Solutions Asia Pte. Ltd. Asia	Diesel Engine Oil DEO SAE 15-W40	15W-40	2	18 l canister: 64247/P 200 l barrel: 65151/D

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Manufacturer & sales region	Product name	SAE grade	Oil category	Material no.
Rolls-Royce Solutions Suzhou Co. Ltd. China	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 64242/P 205 l barrel: 65151/D
	Diesel Engine Oil - DEO 10W-40	10W-40	2	20 l canister: 60606/P
	Diesel Engine Oil - DEO 5W-30	5W-30	3	20 l canister: 60808/P
Rolls-Royce Solutions Asia Pte. Ltd. Indonesia	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 64242/P 205 l barrel: 65151/D
mtu India Pvt. Ltd. India	Diesel Engine Oil - DEO 15W-40	15W-40	2	20 l canister: 63333/P 205 l barrel: 65151/P

Oil change interval

Important

The oil change interval is 1,000 operating hours or max. 1 year under the condition that engine oils of oil category 3 and 3.1 and approved fuels (→ Page 23) are used.
The oil change interval is 500 operating hours or max. 1 year under the condition that engine oils of oil category 2 and 2.1 and approved fuels (→ Page 23) are used.
If fuels which have not been approved are used, shorter oil change intervals are to be expected.
Prior to using non-approved fuels, contact Rolls-Royce Solutions to determine the applicable oil change intervals.

Important

Mixing different engine oils is strictly prohibited!
If, in exceptional cases, the engine oil filled in the engine is not available, top up with another mineral or synthetic engine oil. Ensure that it is approved for mtu products (→ Page 9).
Note the following:

- If you top up an engine oil of lower quality, the maintenance interval corresponding to the lower quality (oil category) must apply. The maintenance interval is reduced.
- If you use an engine with a higher quality, the maintenance interval remains as it is. Observe the specifications in the maintenance booklet.

Changing to another approved oil grade can be done together with an oil change. The remaining oil quantity in the engine oil system is not critical in this regard. This procedure also applies to the genuine Rolls-Royce Solutions engine oil grades in the regions Europe, Middle East, Africa, America and Asia.

Important

When changing to an engine oil of Category 3, note that the improved cleaning effect of these engine oils can result in the loosening of engine contaminants (e.g. carbon deposits).
It may be necessary therefore to reduce the oil change interval and oil filter service life (one time during change).

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2.2 Viscosity grades

Selection of the viscosity grade is based primarily on the ambient temperature at which the engine is to be started and operated. Figure (→ Figure 1) contains guideline values for the temperature limits of the individual viscosity grades.

The temperature specifications of the SAE grade are always based on fresh oils. During operation, engine oil ages due to soot and fuel residue. This results in significant deterioration of the properties of the engine oil particularly at low outside temperatures. At outside temperatures below -20 °C, Rolls-Royce Solutions strongly recommends the use of engine oil of SAE grade 5W-30 or - if approved - 0W-30.

If the prevailing temperature is too low, the engine oil must be preheated.

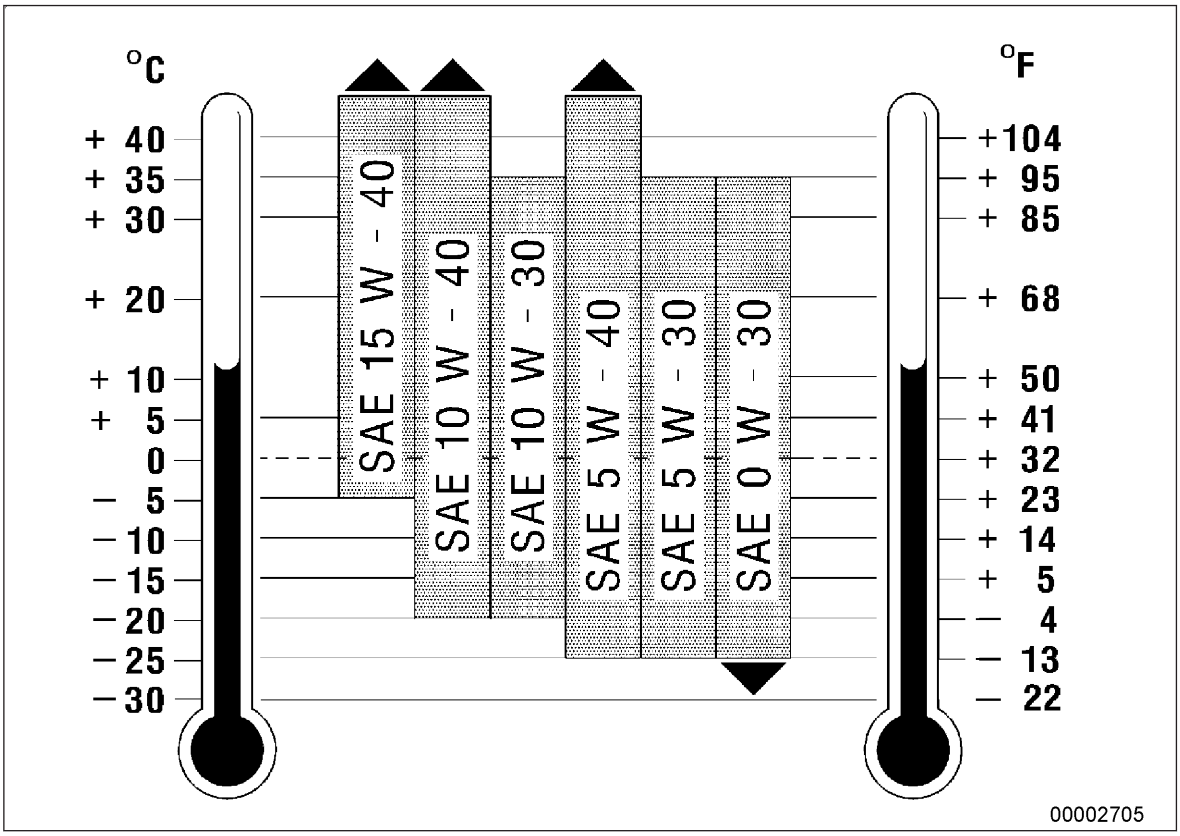

Figure 1: Viscosity grades

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2.3 Series-dependent usability of engine oils

Series-dependent usability of engine oils by oil categories

Series	Approved engine oils	
	Oil categories 2 and 2.1 (Low SAPS)	Oil categories 3 and 3.1 (Low SAPS)
1600Gx0	• Multi-grade oils (→ Page 41) • Multi-grade oils (Low SAPS) (→ Page 50)	• Multi-grade oils (→ Page 54) • Multi-grade oils (Low SAPS) (→ Page 59)

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2.4 Used-oil analysis

In order to check the used oil, it is recommended that regular oil analyses be carried out. Oil samples should be taken and analyzed at least once per year and during each oil change and under certain conditions, depending on application and the engine's operating conditions, sampling / analysis should take place more frequently.

The specified test methods and limit values (Analytical Limit Values for Used Diesel Engine Oils) (→ Table 2) indicate when the results of an individual oil sample analysis are to be regarded as abnormal.

An abnormal result requires immediate investigation and remedy of the abnormality.

The limit values relate to individual oil samples. When these limit values are reached or exceeded, an immediate oil change is necessary. The results of the oil analysis do not necessarily give an indication of the wear status of particular components.

In addition to the analytical limit values, the engine condition, its operating condition and any operational faults are decisive factors with regard to oil changes.

Some of the signs of oil deterioration are:
- Abnormally heavy deposits or precipitates in the engine or engine-mounted parts such as oil filters, centrifugal oil filters or separators, especially in comparison with the previous analysis.
- Abnormal discoloration of components.

Analytical limit values for used diesel engine oils

	Test method	Limit values	
Viscosity at 100 °C max. mm²/s	ASTM D445 DIN 51562 DIN 51569-1	SAE 5W-30 SAE 10W-30	15.0
		SAE 5W-40 SAE 10W-40 SAE 15W-40 SAE 20W-40	19.0
min. mm²/s		SAE 5W-30 SAE 10W-30	9.0
		SAE 5W-40 SAE 10W-40 SAE 15W-40 SAE 20W-40	10.5
Flashpoint °C (COC)	ASTM D92 DIN EN ISO 2592	Min. 190	
Flashpoint °C (PM)	ASTM D93 DIN EN ISO 2719	Min. 140	
Soot content (by weight %)	DIN 51452 CEC-L-82-97	Max. 3.5	
Total base number (mg KOH/g)	ASTM D2896 ISO 3771 DIN 51639	Min. 50% of new-oil value	
Proportion of water (vol. %)	ASTM D6304 EN 12937 ISO 6296	Max. 2000 mg/kg	
Oxidation (A/cm)[1)]	DIN 51453[1)]	Max. 25	

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